

Titanic Survival Analysis Report

1. Executive Summary

The Titanic training dataset contains records for 891 passengers, of whom 342 survived, yielding an overall baseline survival rate of **38.4%**. The analysis shows that survival was primarily dictated by gender, socioeconomic status (ticket class and fare), and family unit size. Passengers adhering to the "women and children first" maritime protocol and those occupying higher accommodation decks had significantly higher odds of evacuation.

Key Metric	Observed Rate / Value	Primary Implication
Overall Survival Rate	38.38% (342 / 891)	Benchmark baseline for all sub-segments
Female Survival Rate	74.20% (233 / 314)	Strongest positive correlation with survival (\$r = +0.54\$)
Male Survival Rate	18.89% (109 / 577)	Lowest survival demographic across adult cohorts
1st Class Survival	62.96% (136 / 216)	Physical proximity to lifeboats and socioeconomic priority
3rd Class Survival	24.24% (119 / 491)	Steepest mortality rate (\$r = -0.34\$ correlation with Pclass)

Optimal Family Unit	57.88% (Small families of 2–4)	Traveling solo (30.4%) or in large groups (16.1%) reduced survival
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2. Data Cleaning & Feature Engineering

- **Missing Values Treated:**
 - **Embarked:** Imputed 2 missing values with the mode ('S').
 - **Cabin:** Converted into **Has_Cabin** (1 for known, 0 for null) and extracted **Deck** initial. Known cabin status exhibited a strong positive correlation ($r = +0.32$) with survival due to deck elevation.
 - **Age:** 177 missing entries were imputed using the median age grouped by **Title** and **Pclass**, preventing distortion caused by a single global median across disparate age cohorts (e.g., child *Master* vs. adult *Mr.*).
- **Feature Engineering:**
 - **Title:** Extracted honorifics from passenger names, grouping rare aristocracy and service ranks into **Royalty** and **Officer**.
 - **AgeGroup:** Segmented continuous age into four operational bins: **Child** (≤ 12), **Teen** (13-19), **Adult** (20-59), and **Senior** (60+).
 - **FamilySize & TravelGroup:** Combined **SibSp** + **Parch** + 1 into **Solo** (1), **Small Family** (2-4), and **Large Family** (5+).

3. Key Findings & Segmented Analysis

Demographic Disparities

- **Gender Priority:** Female survival reached **74.2%**, whereas male survival plummeted to **18.9%**.
- **Age vs. Gender Interaction:** Children under 13 had an above-average survival rate of **57.5%**. However, young boys (titled *Master*) fared dramatically better than adult males (57.5% vs. 15.7%), confirming evacuation priority for young dependents regardless of gender.

Socioeconomic Status

- **Class Hierarchy:** Survival declined strictly down the class ladder: **63.0%** in 1st Class, **47.3%** in 2nd Class, and **24.2%** in 3rd Class.
- **Gender x Class Intersection:** Women in 1st and 2nd Class experienced near-total survival (**96.8%** and **92.1%** respectively). Female survival dropped sharply to **50.0%** in 3rd Class. For men, even 1st Class passengers experienced a modest **36.9%** survival rate, falling to **15.7%** in 2nd Class and **13.5%** in 3rd Class.
- **Ticket Fare:** Survivors paid a median fare of **£26.00** (mean £48.40) versus **£10.50** (mean £22.12) for casualties. Passengers in the highest fare quartile (Q4) survived at **58.1%**, compared to only **19.7%** in the lowest quartile (Q1).

Family Unit Dynamics:

- Traveling solo resulted in a **30.4%** survival rate (537 passengers).
- Small families (2 to 4 members) achieved the highest survival rate at **57.9%**, with families of 4 peaking at **72.4%**.
- Large families (5+ members) suffered high casualty rates, surviving at only **16.1%** (zero survivors recorded among families of 8 or 11), reflecting logistical difficulties in gathering large groups and locating sufficient lifeboat capacity together.

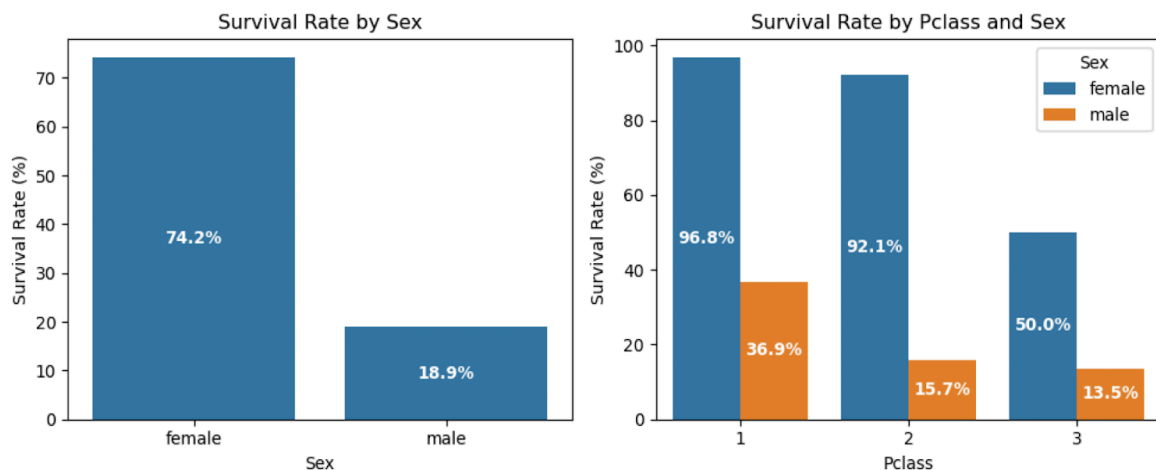
The exploratory analysis confirms that predictive models should prioritize interaction terms between **Sex**, **Pclass**, **FamilySize**, and **Title**. Linear models or tree-based classifiers (such as Random Forest or XGBoost) will benefit directly from these engineered group labels rather than raw passenger attributes.

4. Key Visualizations & Explanatory Analysis

This section presents the primary diagnostic visualizations generated during the analysis, detailing the underlying trends and statistical patterns.

Visualization 1: Demographic & Socioeconomic Survival Disparities

The bar charts below examine overall survival percentages segmented by passenger sex, as well as the interaction between passenger class (**Pclass**) and sex.



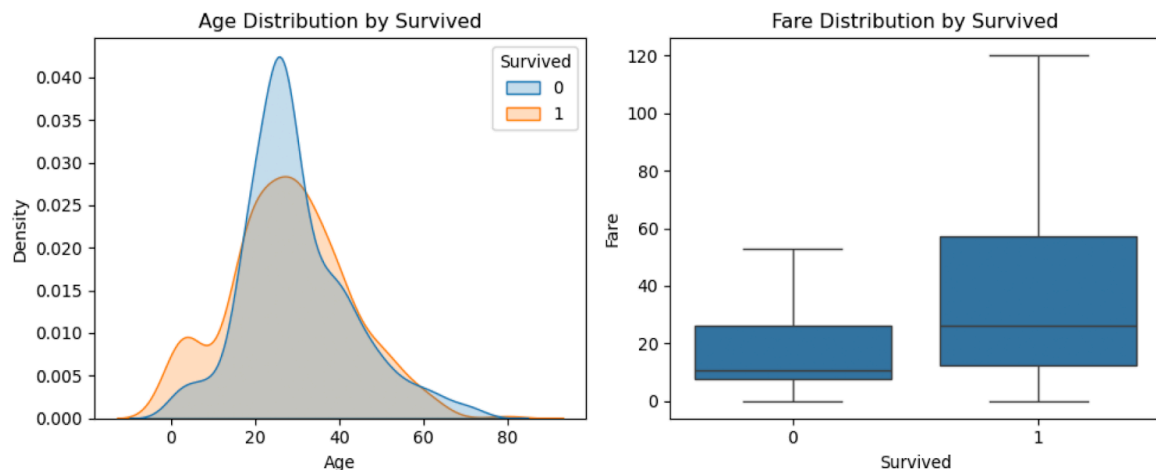
- **Left Subplot (Sex vs. Survival Rate):** Displays the primary gender divide. Female passengers survived at an average rate of **74.2%**, nearly four times higher than male passengers at **18.9%**.
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- **Right Subplot (Pclass & Sex vs. Survival Rate):** Highlights the dual effect of class privilege and gender priority. Female survival in 1st and 2nd class remained near-total at **96.8%** and **92.1%**, before halving to **50.0%** in 3rd class. Men experienced low survival

across all classes, dropping from **36.9%** in 1st class down to **15.7%** in 2nd and **13.5%** in 3rd class.

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Visualization 2: Continuous Feature Distributions (Age & Fare)

The density curve (KDE) and box plot illustrate how passenger age and financial ticket investment affected survival outcomes.



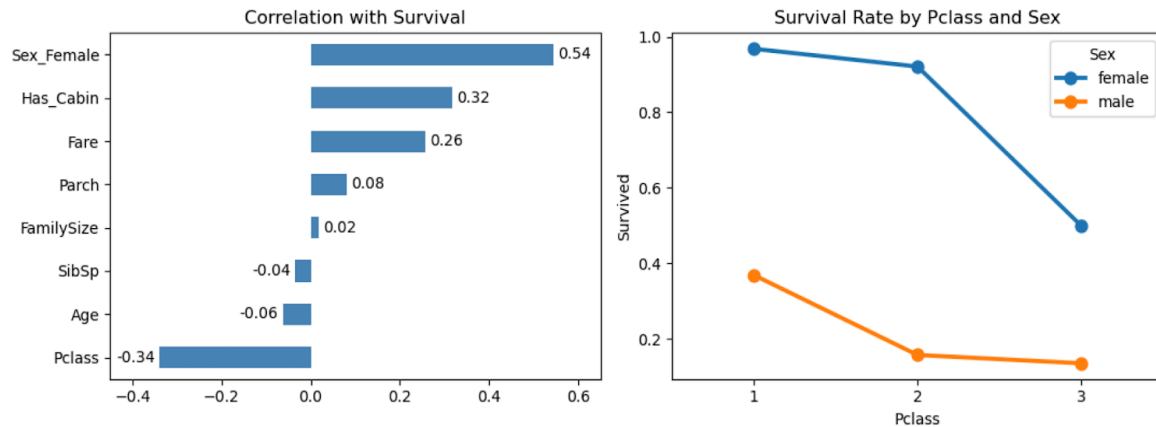
- **Left Subplot (Age Density by Survived):** The prominent bump in the survivor curve (orange, labeled 1) between ages 0 and 10 visually validates the evacuation preference for young children. In contrast, young adults between 18 and 32 years old comprise the sharpest density peak among casualties (blue, labeled 0).
- **Right Subplot (Fare Box Plot by Survived):** Confirms an upward shift in ticket price among survivors. While the median fare for casualties hovers near **£10.50**, survivors show a broader interquartile range with a median fare of **£26.00**, aligning with the safety advantages of upper-deck accommodations.

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Visualization 3: Ranked Feature Associations & Interaction Curves

This panel summarizes the linear relationship of each predictor with the target variable, alongside a direct interaction plot of ticket class by gender.



- Left Subplot (Correlation with Survival):**

- Sex_Female ($r=+0.54$) and Has_Cabin ($r=+0.32$) exhibit the strongest positive correlations with survival, followed by Fare ($r=+0.26$).
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 - Pclass ($r=-0.34$) shows the largest negative correlation, demonstrating that lower-tier accommodations significantly reduced the odds of survival.
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 - Aggregate features like SibSp (-0.04) and overall FamilySize ($+0.02$) demonstrate minimal linear association, reinforcing the non-linear relationship where small families gained an advantage while large groups suffered severe penalties.
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- Right Subplot (Survival Rate by Pclass and Sex Interaction):** Illustrates the interaction dynamics without bar clutter. The steep drop in female survival occurs between 2nd and 3rd class, while the male trajectory experiences its sharpest drop between 1st and 2nd class before plateauing at a low baseline in 3rd class.